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Bioenzyme- A multipurpose enzyme

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Abstract- Bioenzymes work like an 'eco-enzyme'. It is nutrient-dense organic substances obtained by fermentation of vegetables and fruit waste, whose uses are currently being explored in every field. Probiotics and bioactives are abundant in bio-enzymes. As people's consciousness of the need for good and sustainable food systems has risen, the utility of bio-enzymes has also risen. The uses of the bioenzymes in different fields are highlighted in this paper. This is not only the way for waste food management, but it is also useful in other fields like organic cosmetics production, waste-water treatment, the medical field, soil stabilization, detoxification, animal growth, plant growth and seed germination etc. So bioenzyme is a multifunctional enzyme that contributes to nature and human well-being with no side effects.

Keywords: Bioenzyme, Probiotics, Cosmetics production, Germination

INTRODUCTION

Bioenzymes are a kind of protein that is made by fermentation of fruits and vegetable wastes with water and jaggery. It also works as an enzyme to fasten the reaction.[1] It is an organic product and acts as an antimicrobial agent.[2] Permazyme, Renolith, DZ-1X, Fujibeton, Terrazyme- these are the types of bioenzyme.[3] Bioenzymes are very useful in soil stabilization. Alkazyme is also a type of bioenzyme that upgrades the soil's UCS and CBR value.[4] Terrazyme shows the best result for clay stabilization.[5] Different substrates bioenzymes boost the vegetable plant growth height, weight and yield in minimum time.[6] Predominant group of bacteria of phylum- Pseudomonadota (Proteobacteria), class- Betaproteo bacteria, order- Burkholderiales, family- Comamonadaceae found in bioenzyme, which is produced from orange fruit wastes.[7] Multipurpose enzymes like Proteases, Amylases, Cellulases and Lipases are present in bioenzymes.[8] Bioenzymes are used in our daily lives in different ways. It can be used in agricultural area, as natural fertilizers, as water purifier, as natural pesticide, insecticide and herbicide, in nourishment for vegetables and fruits, in Plant growth promoter, maintain to cleanliness, natural health booster for animals, poultry production, antimicrobial, air purification, household cleaning, reduce CO_2, skin and hair care products, laundry, construction of roads, improve soil quality and minimize soil waste.[9] Bioenzyme production is a good way to manage fruit and vegetable wastes.

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USES OF BIOENZYMES

Food Waste Management:-

Bioenzyme formation from food waste is a very effective way to settle-down the dumb waste and overcome the toxicity in the environment and reduce the use of chemicals. It can be prepared with fruit waste, vegetable wastes, kitchen wastes, leaves and other parts of plants. Bioenzymes have compounds for health benefits and also

contain enzymes like lipase, protease and amylase, which help in reducing the carbs, proteins and fats.

Animal Growth and Health-

Adding the bioenzyme to the diet of Pekin ducks affects the growth performance and immunity of poultry. It influences intestinal health at varying dose groups of bioenzyme. Bioenzyme can increase the average daily gain (ADG) and decrease the F: G (feed to gain ratio). In the intestine, bioenzyme reduces the villus height and crypt depth and increases VH/ CD of the jejunum in the low-dose group. It increases the VH, reduces the CD and enhances the VH/ CD of jejunum in the high dose group.[10]

Seed Germination and Plant Growth-

Bioenzyme prepared from kitchen wastes has the ability to enhance the seedling growth and germination of the peanut plant and it can be used as a plant growth promoter by small-holder farmers. Some macro- and micro-nutrients are present in bioenzyme, which helps in the germination of seeds and the growth of plants in soil.[11] Pre-treatment of bioenzyme with supplementation of silica can increase the growth of mung bean seed and yield of mung by increasing pod length and weight. Bioenzyme can improve the quality of mung beans by increasing the protein and carbohydrate content in the yield.[12]



Figure 1: Bioenzyme uses in Agriculture

Detoxification-

Secondary metabolites like Mycotoxins, which are produced by fungi and contaminate the crops. Contaminated crops produce contaminated food, which can be harmful for humans to consume. It affects the immune and nervous system of humans and can also be cancerous. Fungi- contaminated food can be detoxified by bioenzymatic treatment. Most common mycotoxins like Zearalenone, Aflatoxin, Ochratoxin, Deoxynivalenol, Fumonisin and Patulin can be detoxified by bioenzyme.[13] Bioenzymes build up the quality of food and nutritional value, reduce food waste, extend the shelf life of vegetables and improve the food processing.[14]

People are using pesticides to kill pests and to enhance the yield. But pesticides have a bad effect on humans. Food and vegetables should be clean before using. Bioenzyme can be used as a vegetable washing organic liquid. Bioenzyme prepared with Nimba, Tulsi, Lemon peel and Shirisha, which have antioxidant properties, helps in the removal of carcinogenic 'Cypermethrin' pesticide residues from vegetables.[15] Zhang *et al.* (2024)[16] studied that bioenzymes can be used in pesticide detection. Bioenzyme with a combination of nano-enzyme and chromogen has better sensitivity for 'malathion'.

Soil Stabilization-

Chemical fertilizers are expensive and sometimes they become harmful to people. So people turned to cheap, easy-to-use, safe and more beneficial bioenzymes.[17] Terrazyme is a nature-friendly bioenzyme that can be used as a soil stabilizer. It can compact the soil and improve the soil's load-bearing capacity. Bioenzymes re-establish the pH balance of the soil. It manages the proper acidity of soil.[18] Terrazyme -bioenzyme increases the unconfirmed compressive strength (UCS) for soil and becomes a better source for stabilization than chemical soil stabilizers.[19] It boosts the durability of soil stabilization.[20] DZ-IX bioenzyme increases the cohesion in soil particles and stabilizes the black cotton soil.[3]



Figure 2: Bioenzyme use in soil stabilization

Waste Water- Treatment:-

Citrus bioenzyme can detoxify the waste water that is contaminated by electroplating industrial wastes by reducing the BOD, COD, TDS and EC.[21] Bioenzymes help in achieving SDG 6 (Sustainable Development Goal 6).[22] It also helps in making a solar evaporator from Balsa wood, which helps in treating the ammonia nitrogen waste-water. Solar evaporator successfully regulates the capillary action and works as a very effective water transferring channel.[23] Citrus bioenzyme removes the coliform bacteria from domestic waste water.[24]

As Medicine:-

Using the bioenzyme prepared from Mosambi (sweet lime) fruit peel is a cost-effective way to treat the animal's wound. Bioenzyme spray can stop the bleeding more rapidly and heal the wound without side effects. Mosambi bioenzyme has antibiotics like aloin A, quercetin, ononin, isoorientin-2-O-rhamnoside, hesperetin, pterostilbene, 3,7-dirhamnoside and saponarin.[25] Bioenzymes have the potential to treat cancer. Nano-medicines based on bioenzymes increase the therapeutic effectiveness.[26]

Bioenzyme can be used in Hedonic scale odour assessment. Tannic acid, which is present in bioenzyme, helps to cure dental problems with its astringent property.[27]

In Industrial Use:-

Moringa leaves- bioenzyme have Pectinase and Xylanase activity and they can be used in hydro-distillation of mint oil from mentha leaves.[28] Some shampoos and detergents are cancerous and cause many diseases in the human body and are harmful for plant insects due to containing hazardous chemical like Sodium laurel sulphate bioenzyme produced from *Hibiscus rosa sinensis* have conditioning and anti- dandruff properties for hairs, easily detach the oil and dirt from hairs, so it can be used in prepare the shampoo.[29] Bioenzyme produced from citrus can be used in making dishwashing detergents and overcome the health hazards associated with conventional detergents and it is a good cleaner.[30]

CONCLUSION

The necessities of life- land, air, and water are continuously contaminated by industrial development, the growing urbanization and the increasing population day by day. So using the bioenzymes may be an appropriate option for improving food safety. Bioenzymes are good for cleaning, dishwashing, plant growth, cancer treatment and organic cosmetics. In short, this is an organic product that can be used for multiple purposes.

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